

Linguistics 507

Statistical Analysis for Linguists

Spring 2012

Professor: Natasha Warner

Class meets:

8:00-9:15 AM, Tuesday Thursday
Chavez 103

Class website: <http://d2l.arizona.edu>

Office hours: Tu 9:30-10:30, Th. 11-12, Douglass 320

Phone: 626-5591

Email: nwarner@u.arizona.edu

Books/materials:

Keppel, Geoffrey, and Wickens, Thomas. 2004. *Design and Analysis: A Researcher's Handbook*. 4th ed. Prentice Hall.

Readings on correlation and regression from the SPSS manual:

Norušis, Marija J. 2000. *SPSS 10.0 Guide to Data Analysis*. Chapter 19: Linear regression and correlation. pp. 373-406. Prentice Hall.

Norušis, Marija J. 2000. *SPSS 10.0 Guide to Data Analysis*. Chapter 22: Building multiple regression models. pp. 455-488.

Other readings:

Raaijmakers, J.G.W., Schrijnemakers, J.M.C., and Gremmen, F. 1999. How to deal with "The language as fixed effect fallacy": Common misconceptions and alternative solutions. *Journal of Memory and Language*, 41:416-426.

Possibly a reading on Linear Mixed Effects models

Sedlmeier, P. and Gigerenzer, G. 1989. Do studies of statistical power have an effect on the power of studies? *Psychological Bulletin*, 105:309-316.

Keppel is in the bookstore, and the other readings are on the course d2l site (d2l.arizona.edu). The bookstore may shelve Keppel under Psych507, but you don't need any other books Psych507 might require, just Keppel plus the readings on the d2l site.

Optional books you might like:

Takahashi, Shin. 2009. *The Manga Guide to Statistics*. San Francisco: No Starch Press / Ohmsha.

Johnson, Keith. 2008. *Quantitative Methods in Linguistics*. Wiley.

Baayen, R. H. 2008. *Analyzing Linguistic Data. A Practical Introduction to Statistics Using R*. Cambridge University Press.

General organization of the course:

This course will focus on statistical methods that are used in linguistic research, with the most attention given to analysis of variance (ANOVA) and correlation and regression. While a large part of the class will be spent on basics of how to analyze quantitative data, another goal of the

class is to examine the statistical analyses which appear in actual published linguistic literature, and to discuss how students' current and future research might be analyzed statistically.

There will be three homework assignments, a midterm exam, an online "presentation," and a short final paper (**due May 7**). The presentation will be on the statistical analyses used in some published paper. This presentation will be in written form on d2l, and you will also be required to read and comment on others' presentations. The final paper will be on design and analysis of some piece your own research (either already completed, currently in progress, or hypothetical future research). You will also turn in a proposal of the topics for your presentation and paper.

Prerequisites:

The only math you will need will be addition, subtraction, multiplication, and division (just a whole lot of them!). No previous training in statistics is required. The only assumption about background is that you are currently or plan to eventually be involved with quantitative data in some linguistically related field. At this point, you may only be reading literature which uses statistical analyses, or you may already be collecting your own data and need to analyze it.

Requirements/grading:

Homework assignments:	30% (10% each)
Midterm exam:	25%
Topic proposal:	5%
d2l Presentation:	10%
Term paper:	30%

Late policy: 10% reduction of the possible grade per day late, except for documented hospital, police, or similar level excuses.

Tentative schedule:

Date	Topic	Readings/requirements
Week 1 (Jan. 12, 17)	Introduction, why do experiments, why use statistics, areas of linguistics that use statistics, hypothesis testing	Read Keppel Ch. 1
Week 2	1 factor ANOVA with 2 levels, with 3+ levels	Read Keppel Ch. 2-3
Week 3	planned comparisons, assumptions of ANOVA	Read Keppel Ch. 4, 6, 7
Week 4 Lab Day Feb. 7	correction for family-wise error, power, 2 factor ANOVA	Read Keppel Ch. 8, 10-11, Sedlmeier and Gigerenzer
Week 5	interaction, follow-up tests, 3 factor ANOVA	Read Keppel Ch. 12-13 and 21, Assignment 1 due Feb. 14
Week 6	3 factor ANOVA continued, higher designs, review	Read Keppel Ch. 22
Week 7	midterm exam (Feb. 23), ANCOVA	Read Keppel Ch. 15 (after midterm)
Week 8 Lab Day Mar. 1	unequal sample sizes, within-subjects designs	Read Keppel Ch. 14, 16

Week 9	interaction, follow-up tests	Read Keppel Ch. 17, 18; Assignment 2 due Mar. 8
Week 10	Within-subjects continued, mixed designs	Read Keppel Ch. 19-20
Week 11	mixed designs cont'd, random factors, counterbalancing	Read Keppel Ch. 23-24, 26 Topic proposal due Mar. 29
Week 12	by subjects and by items tests	Read Raaijmakers et al
Week 13	correlation and regression	SPSS readings
Week 14, 4/17 podcast (Natasha in DC) Lab Day Apr. 19	multiple regression, signal detection theory, non-parametric tests	SPSS readings Student presentations posted on d2l by Apr. 19
Week 15	Linear Mixed Effects models, cleanup	Read LME reading (if any) Comments due on others' presentations on d2l May 1 Assignment 3 due Apr. 26
May 7, by noon		Paper due

Various statements:

Students with Disabilities:

If you anticipate issues related to the format or requirements of this course, please meet with me. I would like us to discuss ways to ensure your full participation in the course. If you determine that formal, disability-related accommodations are necessary, it is very important that you be registered with Disability Resources (621-3268; drc.arizona.edu) and notify me of your eligibility for reasonable accommodations. We can then plan how best to coordinate your accommodations.

Academic misconduct:

The university's policies about plagiarism, academic honesty, and academic conduct are at <http://dos.web.arizona.edu/uapolicies/>. The library's website also provides extensive help with learning what constitutes plagiarism and how to avoid it. It is particularly important when doing the d2l presentation on a published paper to avoid using the exact words of the paper, except for technical statistical terms. Study groups (especially for homework) are strongly encouraged in this class, but you are not allowed to write up your homework together or share files in the writing process. If you work with a study group, you should make a note on the top of your homework stating who you studied with. Furthermore, if you obtain copies of notes or other materials from students who took this class in past years, you are required to remove any homework or test answer keys from those materials, and to avoid reading them.

Potentially offensive/objectionable content:

The material in this course is primarily not political, but there may be some discussion of endangered languages, use of human subjects in research, dialects associated with various social groups, attitudes toward dialects, etc.

Conduct in class:

Please turn off cell phones when in class. Everyone is required to treat others in class with respect. Disruptive behavior is prohibited.